

scripts/check-cycle-coverage.sh — User Guide

Last verified: 2026-06-26 (§6.AK cycle-coverage gate, against the script at HEAD) **Inheritance:** HelixConstitution §11.4.18 (script documentation mandate) · enforces Lava CLAUDE.md §6.AK clause 1 (Cycle-Coverage Device Gate)

Overview

check-cycle-coverage.sh is the mechanical implementation of the CM-CYCLE-COVERAGE-INTERSECTION gate (§6.AK / §6.AK-debt). It refuses a distribute unless **every CHANGELOG-claimed user-visible fix for the current version has an EXECUTED + PASSED covering device Challenge** in the matching §6.Z evidence file for the **same commit SHA**, recorded within the last 24 hours.

It exists to catch the class of bluff that shipped Lava-Android-1.3.12-1076 (commit 627a0d58): the §6.Z device gate executed ONLY Challenge00 (cold-start) while the CHANGELOG claimed fixes to search, provider-selection, and onboarding — a green gate that proved nothing about the shipped value.

The gate intersects two sets and requires $\text{claim-set} \subseteq \text{executed-pass set}$:

- **CLAIM-SET** — the covering Challenge declared for each CHANGELOG-claimed user-visible fix, read from a per-cycle cycle-coverage-map YAML.
- **EXECUTED-PASS SET** — Challenges that ran and PASSED on a gate-eligible device in the §6.Z evidence file (verdict PASS and runner != host-direct).

Any claim with no covering executed+passed Challenge → reject. It parses YAML and evidence line-by-line with grep/sed/bash (no yq dependency).

Usage

```
# Auto-detect everything (version from app/build.gradle.kts,
# evidence dir from channel,
# map next to the evidence file, HEAD from git, "now" from the
# system clock):
./scripts/check-cycle-coverage.sh --strict

# Explicit version + channel (the form firebase-distribute.sh Gate
# 7 uses):
./scripts/check-cycle-coverage.sh --version="1.3.13-1077" --
channel=debug --strict

# Fully pinned inputs (the form the hermetic test uses – no git,
# no clock, no gradle):
./scripts/check-cycle-coverage.sh --version="9.9.9-9999" --
channel=debug \
--evidence-dir="<dir>" --map="<map.yaml>" \
--head="<sha40>" --now-epoch="<epoch>"
```

Flags

Flag	Default	Meaning
<code>--version</code> <code><name-code></code>	auto-detect from app/ build.gradle.kts (<code>versionName-</code> <code>versionCode</code>)	Version under test, e.g. 1.3.12-1077.
<code>--channel</code> <code><channel></code>	debug	debug or release. Selects the evidence dir when <code>--</code> <code>evidence-dir</code> is not given.
<code>--evidence-</code> <code>dir <path></code>	resolved from channel (see below)	Directory holding <code><version>-test-evidence.</code> <code>{md,json}</code> .
<code>--map</code> <code><path></code>	auto-resolve (see below)	Path to the cycle- coverage-map YAML.
<code>--head</code> <code><sha></code>	git rev-parse HEAD	Expected working-tree HEAD; the evidence commit= must match. Overridable for hermetic tests.
<code>--now-epoch</code> <code><epoch></code>	date +%s	"Now" (Unix epoch) for the ≤24h freshness check. Overridable for determinism.
<code>--strict</code>	on by default (STRICT=1)	Reject on any uncovered claim. The flag is accepted

Flag	Default	Meaning
		but the behavior is the default.
-h, --help	—	Print usage to stderr and exit 0.

Environment overrides (read once at startup, lower precedence than the flags): LAVA_CYCLE_COVERAGE_HEAD (initial value for --head) and LAVA_CYCLE_COVERAGE_NOW_EPOCH (initial value for --now-epoch). An unknown argument prints usage and exits 3.

Path resolution

- **Evidence dir** (when --evidence-dir is omitted): channel release → .lava-ci-evidence/distribute-changelog/firebase-app-distribution; any other channel → .lava-ci-evidence/distribute-changelog/firebase-app-distribution-dev.
- **Evidence file**: the first of <EVIDENCE_DIR>/<version>-test-evidence.md then ...-test-evidence.json that exists (.md preferred). Missing → exit 2.
- **Map** (when --map is omitted): the first of <EVIDENCE_DIR>/<version>-cycle-coverage-map.yaml then .lava-ci-evidence/<version>/cycle-coverage-map.yaml. Missing → exit 2.

Exit codes

Code	Meaning
0	All claims covered by executed + PASSED, same-SHA, ≤24h device rows.
1	One or more claims lack a covering executed+PASSED Challenge (missing / FAIL / SKIP / host-direct), OR the evidence timestamp is older than 24h.
2	Structural: evidence file or map file missing, evidence version= mismatch, or evidence commit= ≠ working-tree HEAD.
3	Internal error: unknown CLI argument, version auto-detect failure, HEAD unresolved, missing/malformed evidence cycle-coverage: header, unparseable timestamp, or a map declaring zero claims.

Input file formats

1. cycle-coverage-map (<vname>--<code>--cycle-coverage-map.yaml)

A small YAML block. The gate reads it line-by-line and only consumes two keys:

```
version: "1.3.12-1077"
claims:
  - bullet: "Search now returns real results"
    covering_challenge: "Challenge58SearchReturnsResults"
  - bullet: "Search filters follow onboarded providers"
    covering_challenge: "Challenge59SearchUsesOnboardedProviders"
```

- Each covering_challenge: value is a **required claim**.
- The preceding bullet: (if any) is used only for human-readable FAIL messages.
- A claim whose covering_challenge is empty/absent is itself an uncovered claim (→ exit 1) — that is the "CHANGELOG claims a fix but no Challenge was written" case the 1076 incident is about.
- A map with **zero** covering_challenge: entries is malformed (→ exit 3).
- Surrounding single/double quotes on values are stripped before matching.

(Other keys such as covers_issues: may appear per the spec but are NOT read by the gate; only covering_challenge: values are matched.)

2. §6.Z evidence file (<vname>--<code>--test-evidence.{md,json})

The §6.Z device-gate evidence file must carry machine-readable result lines the gate consumes (the per-cycle requirement §6.AK adds, written by the cycle author alongside the prose evidence). Two line kinds:

```
cycle-coverage: version=1.3.12-1077 commit=<sha40> channel=debug
timestamp=2026-06-26T12:00:00Z
challenge:
fqnlava.app.challenges.Challenge58SearchReturnsResultsTest
verdict=PASS runner=containers-submodule
```

- Exactly **one** cycle-coverage: header line carries version= / commit= / timestamp= (ISO-8601, e.g. 2026-06-26T12:00:00Z). A missing header, or one missing any of those three fields, is exit 3.
- One challenge: line per executed Challenge, with its verdict= (PASS | FAIL | SKIP) and the device runner=.

- A claim counts as **covered** only when a challenge: line whose fq= **contains** the map's covering_challenge name has verdict=PASS **and** runner != host-direct (host-direct rows are not gate-eligible per §6.AH).
- Matching is substring: the map carries short names (Challenge58Search...), the evidence carries full FQNs (lava.app.challenges.Challenge58Search...Test).

Structural checks on the header: the evidence version= must equal the requested --version (else exit 2); the evidence commit= must match the working-tree HEAD, tolerant of short-vs-long SHAs by prefix compare (else exit 2); the timestamp= must be within 24h of --now-epoch (else exit 1).

How it is wired

- **scripts/firebase-distribute.sh — Phase 1 Gate 7 (§6.AK):** after the §6.P/§6.AA/§6.Z Phase-1 gates, the distribute script invokes this gate for the version+channel being distributed (--evidence-dir="\$CHANGELOG_DIR", --strict) and **refuses the distribute** on a non-zero exit (1 uncovered claim · 2 evidence/map missing/stale/wrong-SHA · 3 internal). It gates both the client and the api-app via the app-resolved changelog dir, with no bypass flag. See docs/scripts/firebase-distribute.sh.md for the Gate 7 detail.
- **.github/pre-push — Check 10 (per spec §5.4):** the spec defines a pre-push Check 10 that runs this gate (--strict) when a pushed commit advances a .lava-ci-evidence/distribute-changelog/<channel>/last-version pointer, so a version-advance (signalling "ready to distribute") cannot be pushed without the covering device evidence.

Constitutional binding (§6.AK clause 1)

No artifact may be distributed unless EVERY CHANGELOG-claimed user-visible fix for the current version has an EXECUTED + PASSED covering device Challenge in the matching §6.Z evidence file for the SAME commit SHA.

The CHANGELOG's user-visible bullets are the claim-set; the gate's executed-PASS Challenge set **MUST** be a superset of the claim-set's covering Challenges. A Challenge00-only (cold-start-only) evidence file **NEVER** satisfies this gate when the cycle claims flow/UI fixes — that is exactly the 1076 incident this gate exists to evict.

Maintenance

When this script is modified, update this document in the same commit (CM-SCRIPT-DOCS-SYNC / .github/hooks/pre-push Check 9 require it). Per §11.4.18 the documentation MUST stay in sync with the codebase. The gate's own hermetic test is tests/cycle-coverage/test_cycle_coverage.sh (spec §5.2 case matrix: positive + six negatives); keep its fixtures in lockstep with any change to the file formats or exit-code contract above.

Cross-references

- scripts/check-cycle-coverage.sh — the script itself
- tests/cycle-coverage/test_cycle_coverage.sh — hermetic falsifiability test
- docs/superpowers/specs/2026-06-26-ak-cycle-coverage-spec.md — the §6.AK spec (gate contract §5.1, test matrix §5.2, hook §5.4)
- docs/scripts/firebase-distribute.sh.md — Phase 1 Gate 7 wiring
- docs/helix-constitution-gates.md — gate inventory
- Lava CLAUDE.md §6.AK (Cycle-Coverage Device Gate) + §6.AK-debt
- HelixConstitution Constitution.md §11.4.18 (script documentation mandate)